A Algorithm

Algorithm 1 outlines the CDCA procedure. Lines 1-5 select target nodes via Eq. 5. We then employ bi-level optimization (lines 6-12), updating the surrogate model in the inner loop and the trigger generator in the outer loop using the joint collateral damage constraints. Finally, lines 13-15 inject the optimized triggers to produce .

Algorithm 1 Algorithm of CDCA

Input: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}), \mathcal{Y}_L, y_t, \beta, \gamma$.
Output: Backdoored graph $\mathcal{G}'$, adaptive trigger generator f_g.

1: Initialize $\mathcal{G}' = \mathcal{G}$;
2: Randomly initialize θ_g for f_g and θ_s for f_s;
3: Select target nodes $\mathcal{V}_p$ based on Eq. 5;
4: Identify neighbor set $\mathcal{V}_n$ and obtain pseudo-labels y' via trained f_s;
5: Construct Target Manifold $\mathcal{C}'$ and Clean Manifold $\mathcal{C}$ based on y';
6: **while** not converged yet **do**
7:     **for** $t = 1, 2, ..., N$ **do**
8:         Update θ_s by descent on $\nabla_{\theta_s} \mathcal{L}_t$ based on Eq. 11;
9:     **end for**
10:     Update θ_g by descent on $\nabla_{\theta_g} (\mathcal{L}_t + \beta \mathcal{L}_c + \gamma \mathcal{L}_s)$ based on Eq. 12;
11: **end while**
12: **for** $v_i \in \mathcal{V}_p$ **do**
13:     Generate the specific trigger g_i for v_i by using f_g;
14:     Update $\mathcal{G}'$;
15: **end for**
16: **return** $\mathcal{G}'$ and f_g;

B Time Complexity Analysis

The time complexity of CDCA primarily stems from two phases: node selection and bi-level trigger generation.

Node Selection. This phase involves computing node embeddings using a pre-trained benign GCN and calculating selection scores. The inference cost for an L-layer GCN is $\mathcal{O}(L|\mathcal{E}|d)$, where $|\mathcal{E}|$ is the number of edges and d is the feature dimension. Computing the centrality and ranking the scores takes $\mathcal{O}(|\mathcal{V}| \log |\mathcal{V}|)$, where $|\mathcal{V}|$ is the number of nodes. Since $|\mathcal{V}| \log |\mathcal{V}|$ is typically dominated by the message-passing cost in large graphs, the complexity for this phase is approx $\mathcal{O}(L|\mathcal{E}|d)$.

Bi-level Trigger Optimization. The training process involves an iterative bi-level optimization loop. Let T denote the total number of outer iterations and N be the number of inner steps for the surrogate model update.

- *Surrogate Model Update (Inner Loop):* In each inner step, updating the surrogate GCN requires standard forward and backward passes, with a complexity of $\mathcal{O}(L|\mathcal{E}|d)$. For N steps, this amounts to $\mathcal{O}(N \cdot L|\mathcal{E}|d)$.
- *Trigger Generator Update (Outer Loop):* This step involves trigger generation and neighbor contrastive loss computation. Since these operations are confined to local subgraphs, their computational cost is negligible relative to the full graph propagation. Consequently, the complexity of this step remains dominated by the GCN forward pass $\mathcal{O}(L|\mathcal{E}|d)$.

Attack	Model	CA	PAC	ASR	ASR-2
GTA	GCN	84.93	81.16	94.57	12.13
	GAT	84.07	79.50	85.59	11.20
	GraphSAGE	83.76	80.49	90.67	7.04
UGBA	GCN	84.83	80.65	94.21	13.06
	GAT	76.67	74.78	92.21	39.45
	GraphSAGE	84.07	82.62	94.86	9.68
DPGBA	GCN	84.91	81.02	97.04	22.21
	GAT	83.70	73.07	97.17	22.52
	GraphSAGE	82.22	82.32	97.72	22.44
SPEAR	GCN	84.98	48.15	95.89	80.07
	GAT	82.51	53.34	99.42	57.52
	GraphSAGE	84.21	66.79	100.0	38.51
CDCA	GCN	84.98	84.98	98.55	1.77
	GAT	83.70	85.95	99.76	1.60
	GraphSAGE	84.28	85.89	100.0	1.57

Table 4: Backdoor attack performance (%) on PubMed across different GNNs architectures.

Overall Complexity. Aggregating these components, the total time complexity of CDCA is $\mathcal{O}(T \cdot (N+1) \cdot L|\mathcal{E}|d)$. Crucially, this complexity is linear with respect to the number of edges $|\mathcal{E}|$, which is asymptotically consistent with standard GNN training. This indicates that CDCA is computationally efficient and scalable to large-scale graph datasets.

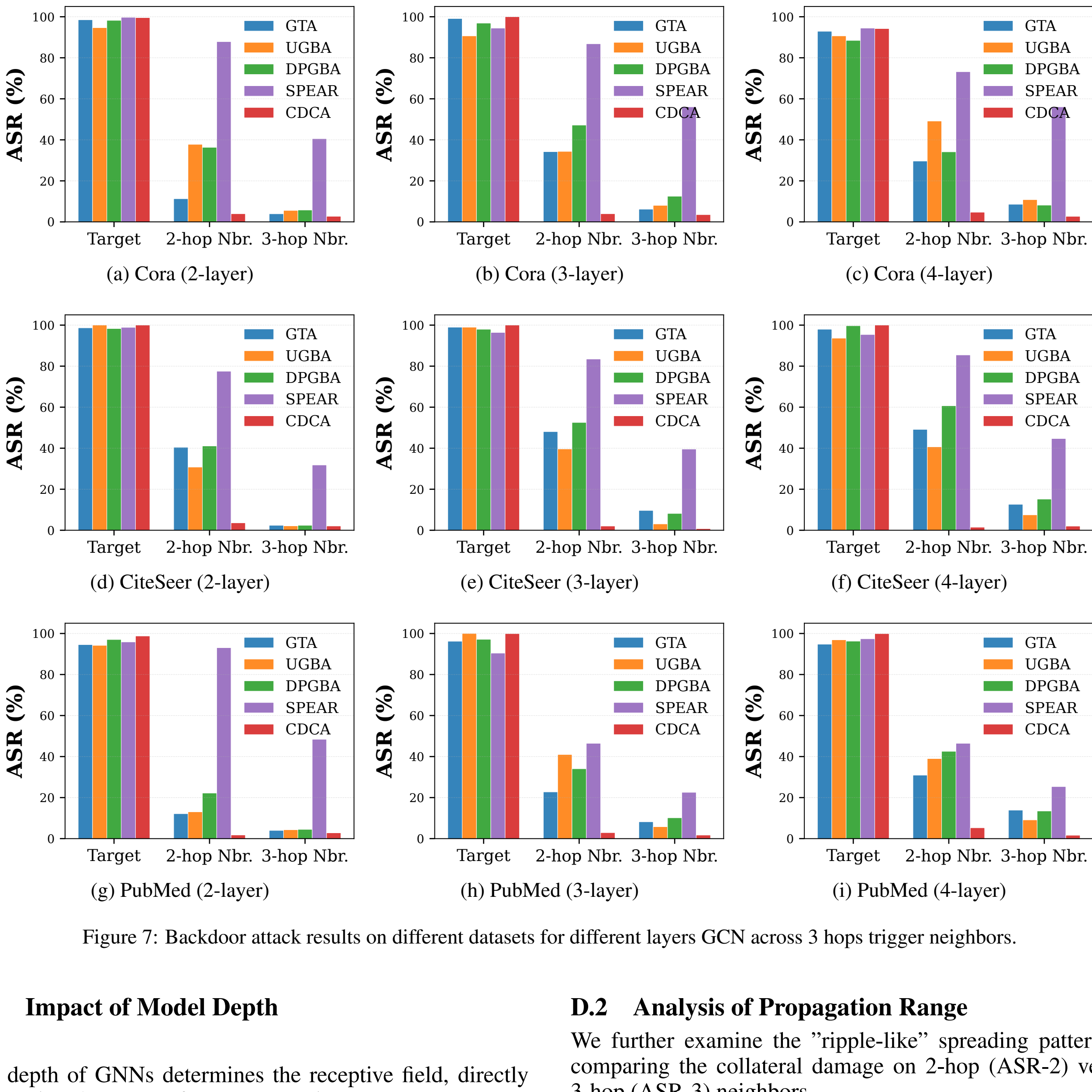
C Detailed Experiments on Attack Transferability

In addition, to examine the transferability of backdoor attacks and their impact on collateral damage using various test models, we conduct experiments on mainstream models with 2-layer. Specifically, we evaluate transferability of backdoor attacks by using GCN [Kipf and Welling, 2017], GAT [Veličković *et al.*, 2018] and GraphSAGE [Hamilton *et al.*, 2017] as victim models. The results are shown in Table 4, where we can observe that:

- The evaluation reveals significant collateral damage across all frameworks, albeit to varying degrees, with GraphSAGE exhibiting lower CDR. This may stem from its aggregation method, which facilitates a more localized information flow and mitigates the risk of widespread misclassification among neighboring nodes.
- CDCA shows promising results for both high ASR and minimal CDR, demonstrating its effectiveness and transferability across different GNNs architectures.

D Detailed Experiments on Attack Performance

To evaluate the robustness of CDCA, we extend our analysis to varying GNN model depths and multi-hop propagation ranges, as illustrated in Fig. 7.



(a) Cora (2-layer) (b) Cora (3-layer) (c) Cora (4-layer)

(d) CiteSeer (2-layer) (e) CiteSeer (3-layer) (f) CiteSeer (4-layer)

(g) PubMed (2-layer) (h) PubMed (3-layer) (i) PubMed (4-layer)

Figure 7: Backdoor attack results on different datasets for different layers GCN across 3 hops trigger neighbors.

D.1 Impact of Model Depth

The depth of GNNs determines the receptive field, directly influencing the scope of message passing.

Baselines suffer from deeper propagation. As the number of layers increases (from 2 to 4), baseline methods exhibit a sharp rise in collateral damage. This trend suggests that with expanded receptive fields, unconstrained triggers in baselines inevitably contaminate a larger set of neighbors, rendering the attack significantly less stealthy.

Stability of CDCA. In contrast, CDCA demonstrates remarkable stability across varying model depths. It consistently maintains low collateral damage even in deeper architectures, significantly outperforming baselines like SPEAR and UGBA. Meanwhile, CDCA sustains a high target Attack Success Rate (ASR-1) across all depths, proving that our neighborhood contrastive constraint effectively localizes the backdoor effect without compromising attack potency in deeper networks.

D.2 Analysis of Propagation Range

We further examine the "ripple-like" spreading pattern by comparing the collateral damage on 2-hop (ASR-2) versus 3-hop (ASR-3) neighbors.

Decay and Residual Damage. While malicious influence naturally decays as the hop distance increases (i.e., ASR-3 is consistently lower than ASR-2), baseline attacks retain substantial residual damage, particularly in deeper networks. In 4-layer models, the collateral damage at the 3-hop level remains alarmingly high, indicating that unconstrained triggers generate strong noise that pollutes a significant portion of the graph.

Effective Containment. CDCA not only minimizes immediate collateral damage but also effectively cuts off long-range propagation. Across all datasets, CDCA suppresses ASR-3 to negligible levels. These results confirm that CDCA successfully confines the backdoor function to the intended target vicinity, preventing the spillover of malicious signals to the broader graph structure.